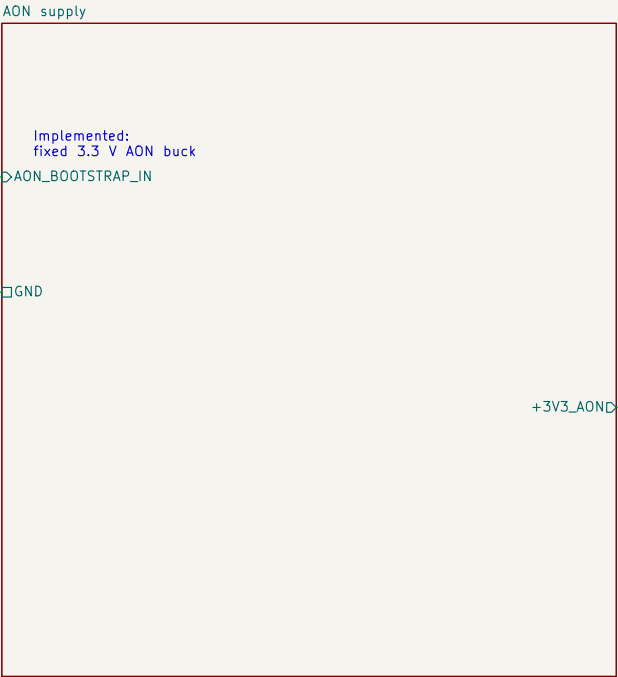
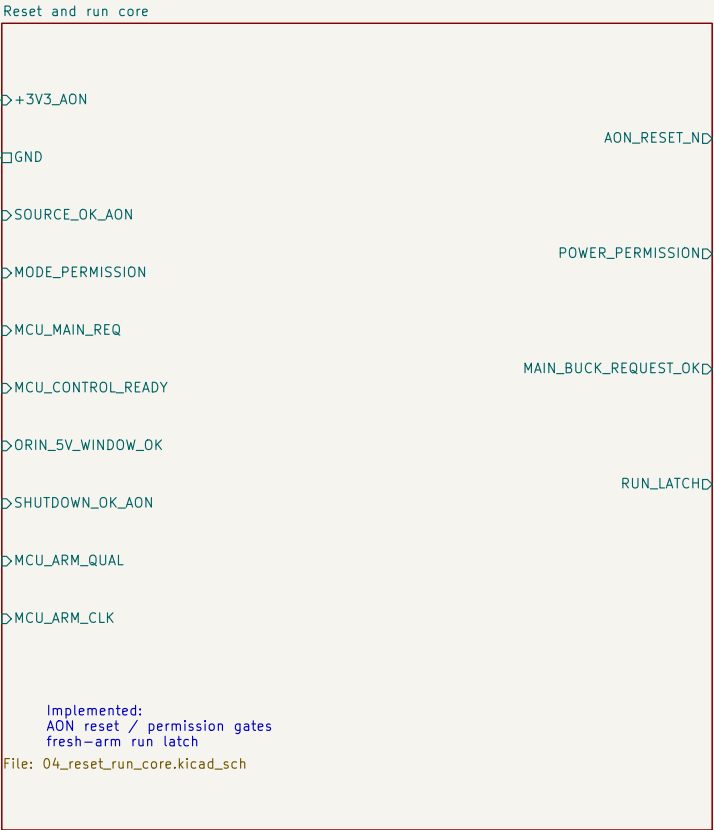


EMBER PAYLOAD COMPUTE – NEXT PROTOTYPE / STAGE 2



File: 03_aon_supply.kicad_sch



File: 02_bootstrap_inputs.kicad_sch

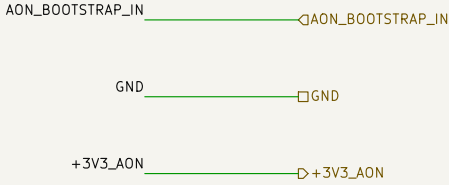
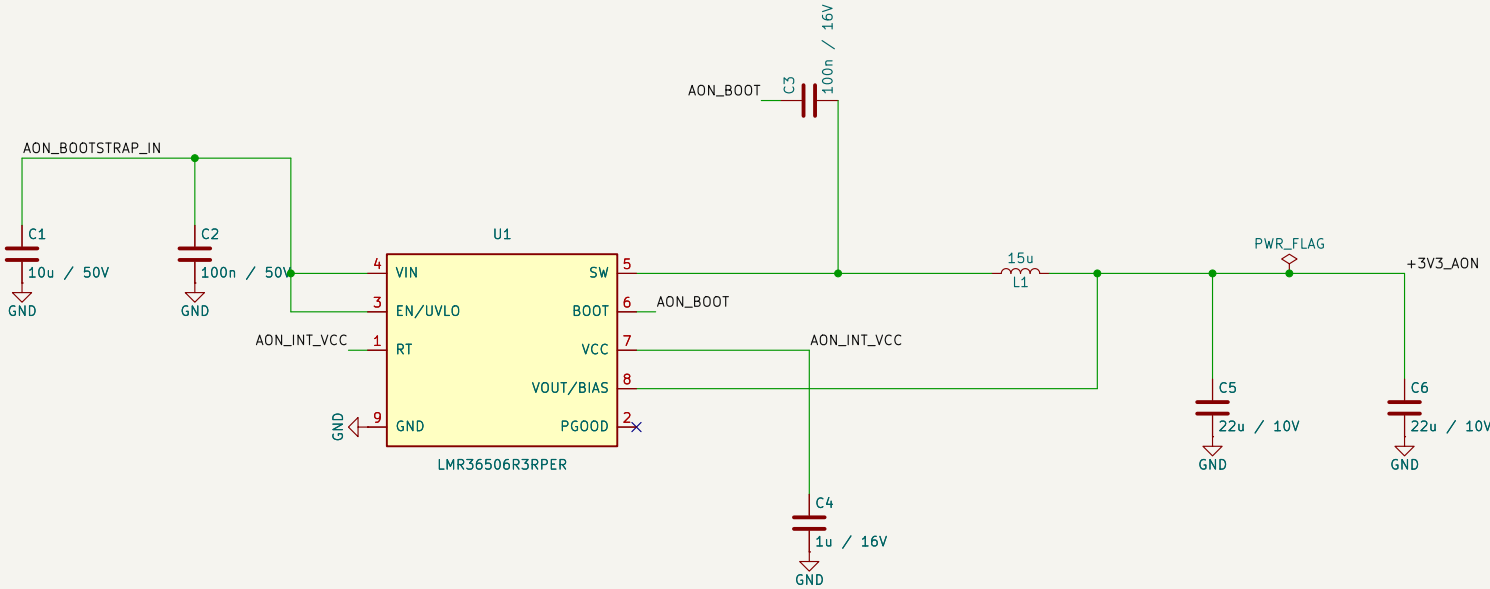
BOUNDARIES OPEN / FAIL-OFF DEFAULTS
SOURCE_OK_AON: immutable source and qualified raw windows.
MODE_PERMISSION: sampled bench / live IHU permission.
MCU_CONTROL_READY: external watchdog-qualified readiness.
MCU request / arm: supervisor and firmware pending.
ORIN_5V_WINDOW_OK / SHUTDOWN_OK_AON: qualified receivers.
MAIN_BUCK_REQUEST_OK: logic request; isolated EN driver pending.
RUN_LATCH: logic permission; Jetson POWER_EN interfaces pending.

STAGED CAPTURE: NOT LAYOUT, FAB OR FLIGHT RELEASE
AON0.200W startup target /0.250W ceiling; cold-ramp tests pending.
C-to-C initial>=500mA required; full USB target15V/3A.
Conservative7W bring-up;15W protection qualification remains open.
No stack +3V3/+5V joins AON. Main rails/source lock remain future.
Entry faults, flight inhibit, PD, connectors and PCB still unimplemented.

Sheet: /
File: payload_compute_rev_next.kicad_sch

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KiCad E.D.A. 10.0.0		Id: 4/4

03 – ALWAYS-ON SUPPLY / IMPLEMENTED CORE



RT tied to internal VCC; nominal 1 MHz. VCC supplies only RT selection and C4; no external load.
15 uH, 2 x 22 uF output are nominal prototype values; effective C / saturation / ripple / MPN remain to qualify.
PGOOD unused. AON reset is supervised separately by TPS3808G33. EN follows protected bootstrap feed.

INTERFACE / ASSEMBLY GATES
AON_BOOTSTRAP_IN is NOT raw stack or unprotected USB. Independent limited reverse-isolated bootstrap feeds are pending.
Chosen fixed-output DC allocation: 3.2505–3.3825 V; max G33 release 3.1928 V; static margin 57.7 mV.
Startup budget target 0.200 W (old ceiling 0.250 W); transients, lower-feed startup and attach current unmeasured.
PWR_FLAG on generated +3V3_AON declares the real buck output through L1; no flag declares missing input/return.
Accepted RPE prototype lands differ from TI contour; common 0.100 / 0.125 mm stencil still to reconcile.

04 - AON RESET / PERMISSION / FRESH-ARM RUN CORE

POWER_PERMISSION = AON_RESET_N & SOURCE_OK_AON & MODE_PERMISSION
MAIN_BUCK_REQUEST_OK = POWER_PERMISSION & MCU_MAIN_REQ & MCU_CONTROL_READY
RUN_CLEAR_N = MAIN_BUCK_REQUEST_OK & ORIN_5V_WINDOW_OK & SHUTDOWN_OK_AON
RUN_LATCH samples MCU_ARM_QUAL on rising MCU_ARM_CLK only while RUN_CLEAR_N is high.
Fault/recovery does not generate an arm edge. PRE_N tied high; Q_N and TPS3808 CT intentionally NC.

BOUNDARY / TIMING CONTRACT
All external high-true qualifiers and MCU controls have 10k pulldowns (about 0.342 mA worst at 3.3825 V / 9.9k).
Future drivers must meet LVC VIH >= 2.0 V / VIL <= 0.8 V with these loads, ramps and partial power checked.
CT open: AON release 12-28 ms; not source/reference/module readiness. MR_N tied high.
Arm: keep CLK low, set D, wait >= 1 ms, pulse CLK >= 1 ms, lower CLK then D. Race/recovery tests remain.
Logic request is not physical buck EN. RUN_LATCH is not direct 5 V Jetson POWER_EN.

Sheet: /Reset and run core/ File: 04_reset_run_core.kicad_sch		
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Size: A3	Date:	Rev:
KiCad E.D.A. 10.0.0		Id: 3/4

POWER_PERMISSION = AON_RESET_N & SOURCE_OK_AON & MODE_PERMISSION
MAIN_BUCK_REQUEST_OK = POWER_PERMISSION & MCU_MAIN_REQ & MCU_CONTROL_READY
RUN_CLEAR_N = MAIN_BUCK_REQUEST_OK & ORIN_5V_WINDOW_OK & SHUTDOWN_OK_AON
RUN_FAULT samples MCU_ARM_EQUAL on rising MCU_ARM_CLK only while RUN_CLEAR_N is high.
Latch/recovery does not generate an arm edge. PRE_N tied high; Q_N and TPS3808 CT intentionally NC.

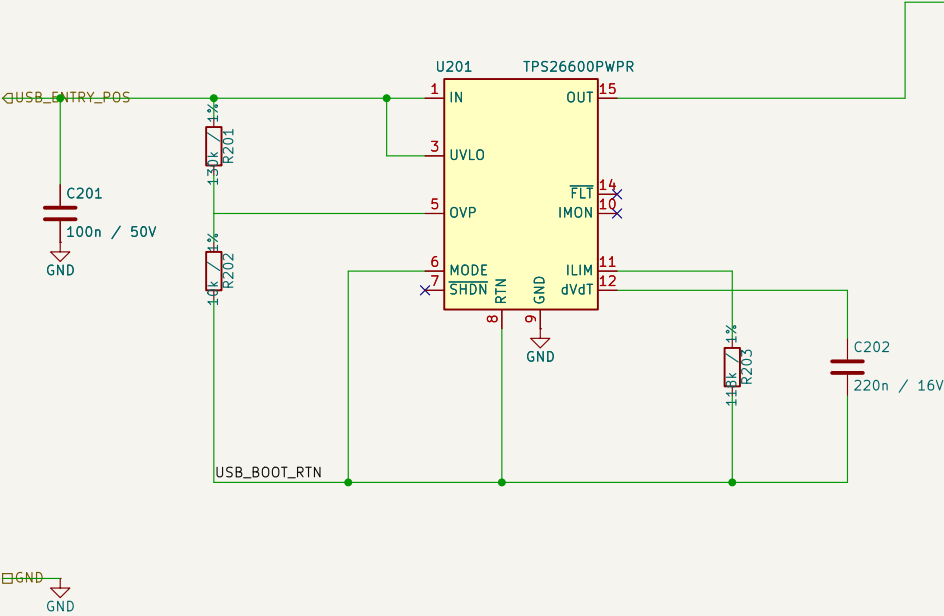
BOUNDARY / TIMING CONTRACT

All external high-true qualifiers and MCU controls have 10k pulldowns (about 0.342 mA worst at 3.3825 V / 9.9k).
 Future drivers must meet LVH VIH >= 2.0 V / VIL <= 0.8 V with these loads, ramps and partial power checked.
 CT open: AON release 12-28 ms; not source/reference/module readiness. MR_N tied high.
 Arm: keep CLK low, set D, wait >= 1 ms, pulse CLK >= 1 ms, lower CLK then D. Race/recovery tests remain.
 Logic request is not physical buck EN. RUN_LATCH is not direct 5 V Jetson POWER_EN.

Sheet: /Reset and run core/ File: 04_reset_run_core.kicad_sch		
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KiCad E.D.A. 10.0.0		Id: 3/4

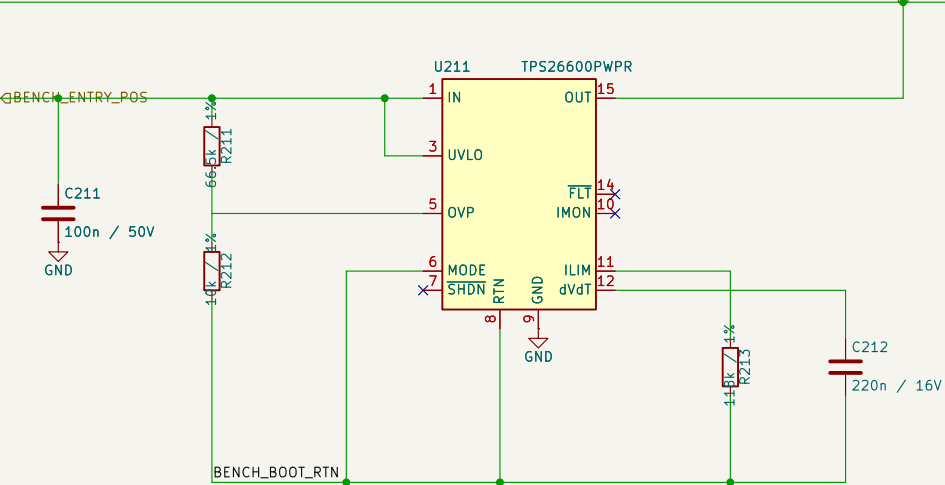
USB autonomous bootstrap

C-to-C >=500mA at initial5V
5V attach / 15V PD operating



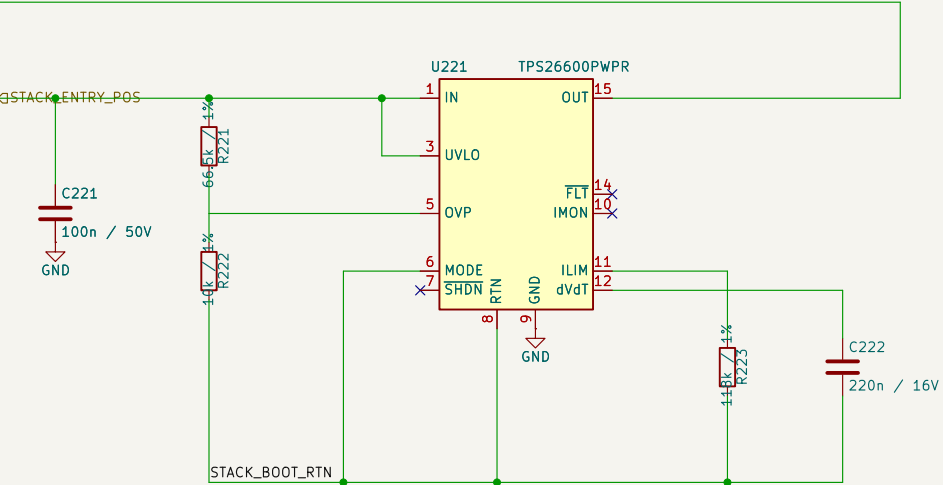
BENCH autonomous bootstrap

6-8.4V normal source
Entry fault envelope provisional



STACK autonomous bootstrap

6-8.4V normal source
Entry fault envelope provisional



Stage2: autonomous limited AON paths; main admissions still absent.
USB UVLO is tied directly to IN to allow initial5V, not factory15V.
SHDN self-biased open; MODE=localRTN gives active limit/auto-retry.
All three RTN/EP domains remain separate from GND and one another.
Native reverse-blocking OR replaces Schottky; leakage/transients need tests.
RILIM118k/1% stays below120k recommendation; short limit ~123mA max.
Cin100nF each; shared switched AON input has10uF+100nF on supply sheet.
Reduced AON0.200W target /0.250W ceiling; CAN standby until full profile.
Source connector, PD, coordinated entry fuse/TVS and flight inhibit are boundaries.
Inputs provisional<=24V fault envelope; no60V system rating claimed.
Cold attach, slow ramp with VIN-enabled buck, restart/leakage remain scope gates.

CdVdt220nF / 20% allocation; output slew ~360-797V/s.
Charging12.12uF switched input capacitance needs <=9.66mA at ramp end.
This arithmetic does not bound initial buck/output-cap charging load.
Bootstrap OR is independent of main immutable source selection.
Fault retry or complete AON loss must never synthesize a new arm.